Converting Cadence ViVa Plots to TikZ pictures for LaTeX

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Introduction

Virtuoso simulation suite from Cadence Design Systems is used extensively for analog circuit design. The quality of the plots from Virtuoso Visualization and Analysis (ViVa), the built-in visualization tool is not of sufficient quality for publishing. This skill script converts the selected waveforms in a given subWindow of ViVa to a TikZ picture or a pdf file for inclusion in LaTeX documents. TikZ picture can be generated solely using skill. Generating a pdf file requires a local LaTeX installation.

Ideally, I would have liked to arrange the waveforms in ViVa and export them to TikZ with the same strips, ordering of waveforms, axis limits, axis labels, and legends. Unfortunately, I could not find a way to get all this information using skill scripts. Therefore, this script exports the selected waveforms from a subwindow into a single axis or a set of strips arranged in a single column[1] In the latter case, each strip has a single waveform. If a multi-strip plot with more than one waveform is desired in a single strip, a two-step process is necessary. Axis limits, legends and some other options can be set using this script. Any further customization can be carried out in the TikZ picture. The script outputs a file that can be included in the LaTeX document.

[1]The single row version currently has weird bugs and is disabled.

Usage

Load the skill file using the following in the CIW. You have to specify the complete path if the skill file is in a different folder than where you executed "virtuoso".

load("nkWaveToTikz.il")

Select the waveforms and run the following in the CIW. All arguments are optional. Arguments can be in any order, but must be in the keyword form, i.e. the keyword `?keyName` followed by the argument `_keyName`.

```
nkWaveToTikz(
?fileName _fileName
?plotMode _plotMode
?width _width
?relheight _relheight
?traceWidth _traceWidth
?precision _precision
?xLimits _xLimits
?yLimits _yLimits
?xLabels _xLabels
?yLabels _yLabels
?xTickLabels _xTickLabels
?yTickLabels _yTickLabels
?xTicks _xTicks
?yTicks _yTicks
?colors _colors
?legends _legends
?xModes _xModes
?yModes _yModes
?textLabels _textLabels
?minorGrid _minorGrid
?markers _markers
?waveAssign _waveAssign
?undersample _undersample
?stripVsep _stripVsep
?noAxes _noAxes
?genPdf _genPdf
?retainTexFile _retainTexFile
?headerFilePath _headerFilePath
?additionalAxisArgs _additionalAxisArgs
)
```

Arguments

Argument	Description	Default behavior
_fileName	File name	Date in the UNIX format without spaces. e.g. Nov132220222023.tikz
_plotMode	single: All selected curves on the same axes column: Strips in a single column row: Strips in a single row; Currently not functional; Gives the same result as column	single
_width	Figure width (no real need to change this; Can be scaled in the LaTeX document)	\textwidth
_relheight	Height relative to width	1
_traceWidth		"thick"
_xLimits	List† with x-axis limits. e.g., '((0 100) (0 50))	Auto set by tikZ to fit the data
_yLimits	List† with y-axis limits. e.g., '((0 1e-5) (0 1))	Auto set by tikZ to fit the data
_xLabels	List† with x-axis label string	No label
_yLabels	List† with x-axis label string	No label
_xTickLabels	List with 1 or 0 to enable or disable the numbers next to x-axis ticks. e.g., (1 0 0). Must have as many elements as the number of strips†.	Numbers next to ticks enabled Auto set by tikZ
_yTickLabels	List with 1 or 0 to enable or disable the numbers next to y-axis ticks. e.g., (1 0 0). Must have as many elements as the number of strips†.	Numbers next to ticks enabled
_xTicks	List† of lists (as many as strips) of numbers at which x-axis ticks must appear.	Auto set by tikZ
_yTicks	List† of lists (as many as strips) of numbers at which y-axis ticks must appear.	Auto set by tikZ
_colors	List‡ of colors (as many as waveforms)	Default color order of pdf/TikZ
_legends	List of lists (as many as strips) of strings which will be the waveform legends	
_xModes	List with as many elements as there are strips. Each must be "lin" or "log".	"lin"
_yModes	List with as many elements as there are strips. Each must be "lin" or "log".	"lin"
_textLabels	List of lists of text labels; List of list (of labels on each strip) of lists (of label details). e.g. '(((strip1 label1) (strip1 label2)) '(strip2 label1)) ; (strip1 label1) = (x y text) where x and y are the coordinates and text (without quotes) is the text label	
_minorGrid	List with as many two-element lists as there are strips. Each two-element lists corresponds to the number of minor ticks between major ticks for x and y axes respectively.	"major"
_markers	0: No markers 1: Marker at every point $n > 1$: Marker at every nth point	0: No markers
_waveAssign	List with as many lists as there are strips. Each list must contain the list of waveforms that must be in that strip. Required for having multiple strips with more than one waveform in a single strip.	All waveforms on the same axes or each waveform on separate axes. (depending on plotmode)
_undersample	Positive integer n to sample every nth point Undersample the waveform to reduce the number of points in the tikz plot	$n = 1$
_stripVsep	String to specify additional vertical spacing between strips Standard LaTeX spacing, e.g., "1cm"	
_noAxes	No axes boxes; Useful to get only the waveshape e.g., use in block diagrams	With axes boxes
_genPdf	Generate pdf. Needs the headerFile. See the "Notes" section.	No
_headerFilePath	Path to the header file	Virtuoso working directory
_retainTexFile	The intermediate LaTeX-related files used for pdf generation are deleted. To retain the LaTeX file for debugging the pdf output, set this to 1.	Delete LaTeX-related files
_additionalAxisArgs	Additional arguments to \\begin{axis}[]; List with one per strip; These will be placed at the end and will supersede any options set earlier '("axis lines=center") will make axis lines at the center instead of left '(nil "axis lines=center") will make axis lines at the center for the second strip	

† _xLimits, _yLimits, _xLabels, _yLabels, _xTickLabels, _yTickLabels, _xTicks, _yTicks should be lists with as many items as the number of strips. If the list length is less than the number of strips, the last one will be applied to the unassigned strips. To have a common specification for all the strips, a list with a single item can be used.

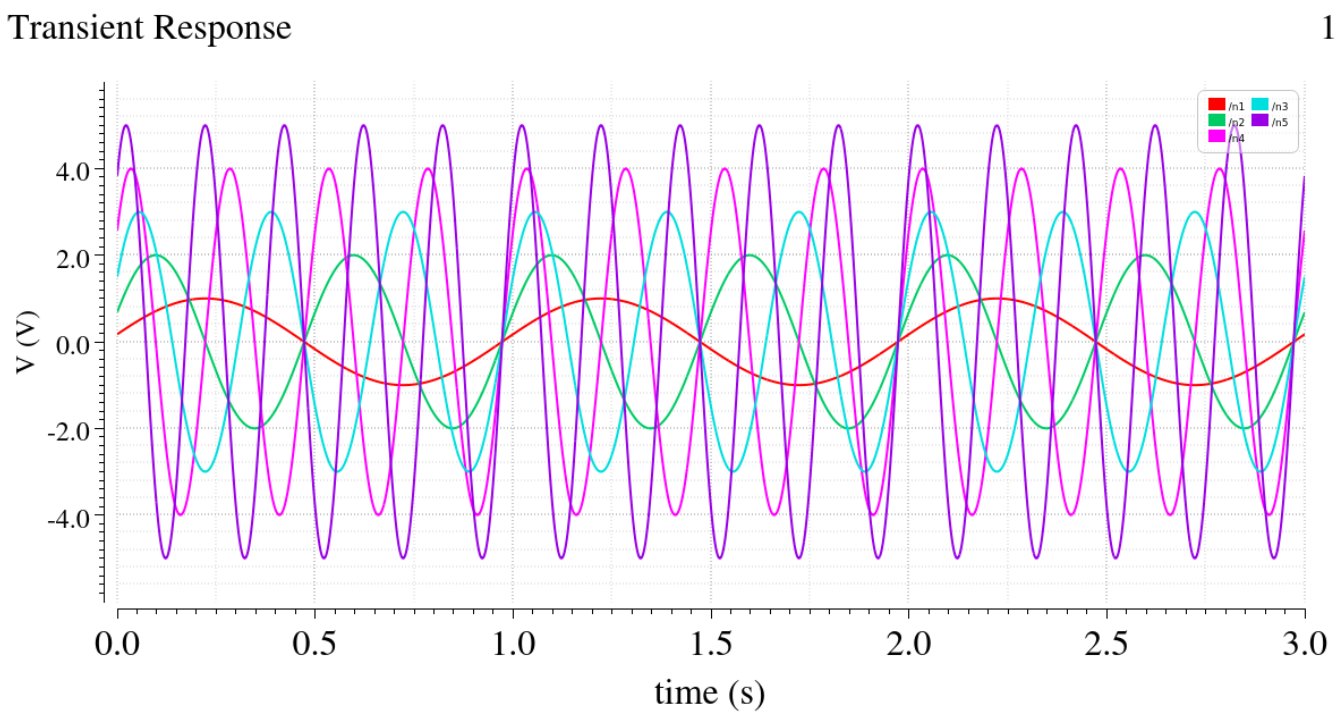
‡ If the list has fewer entries than the number or waves, default color ordering is used for the remaining ones.

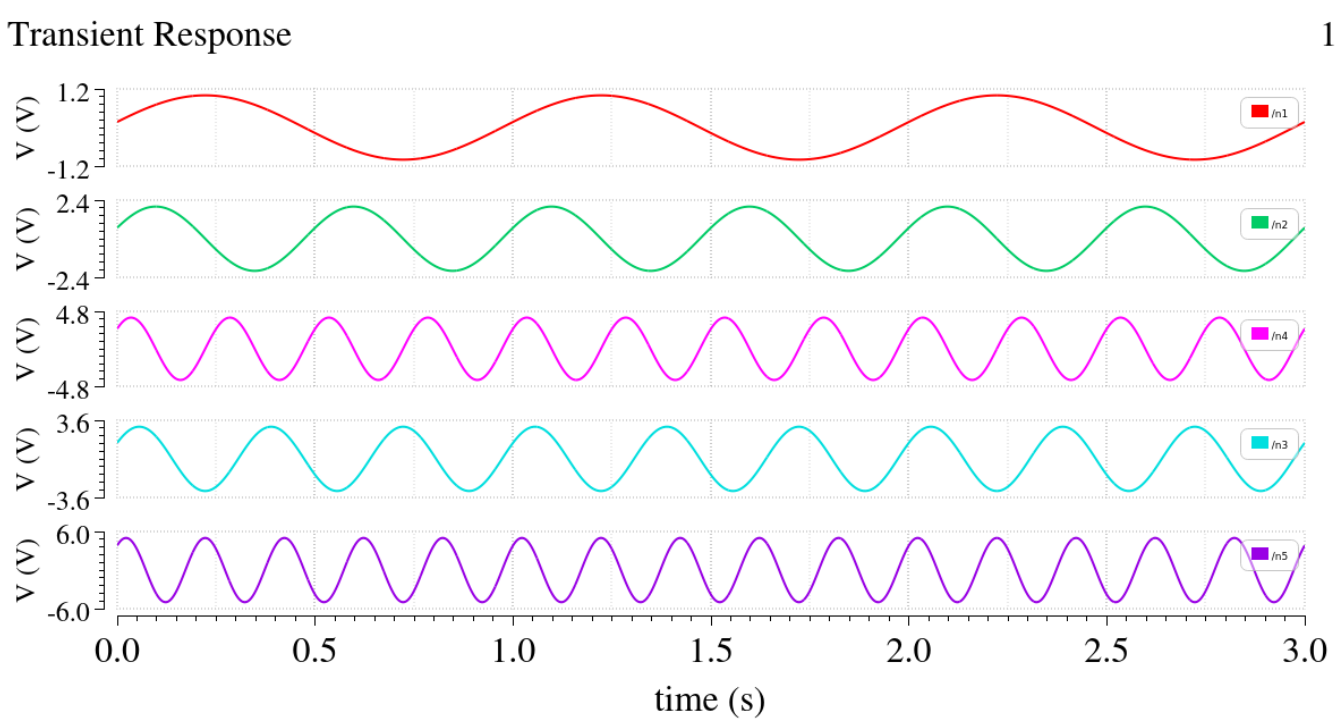
Use nil (without quotes) as a list element to ignore it for that wave or strip.

Examples

Plots from ViVa

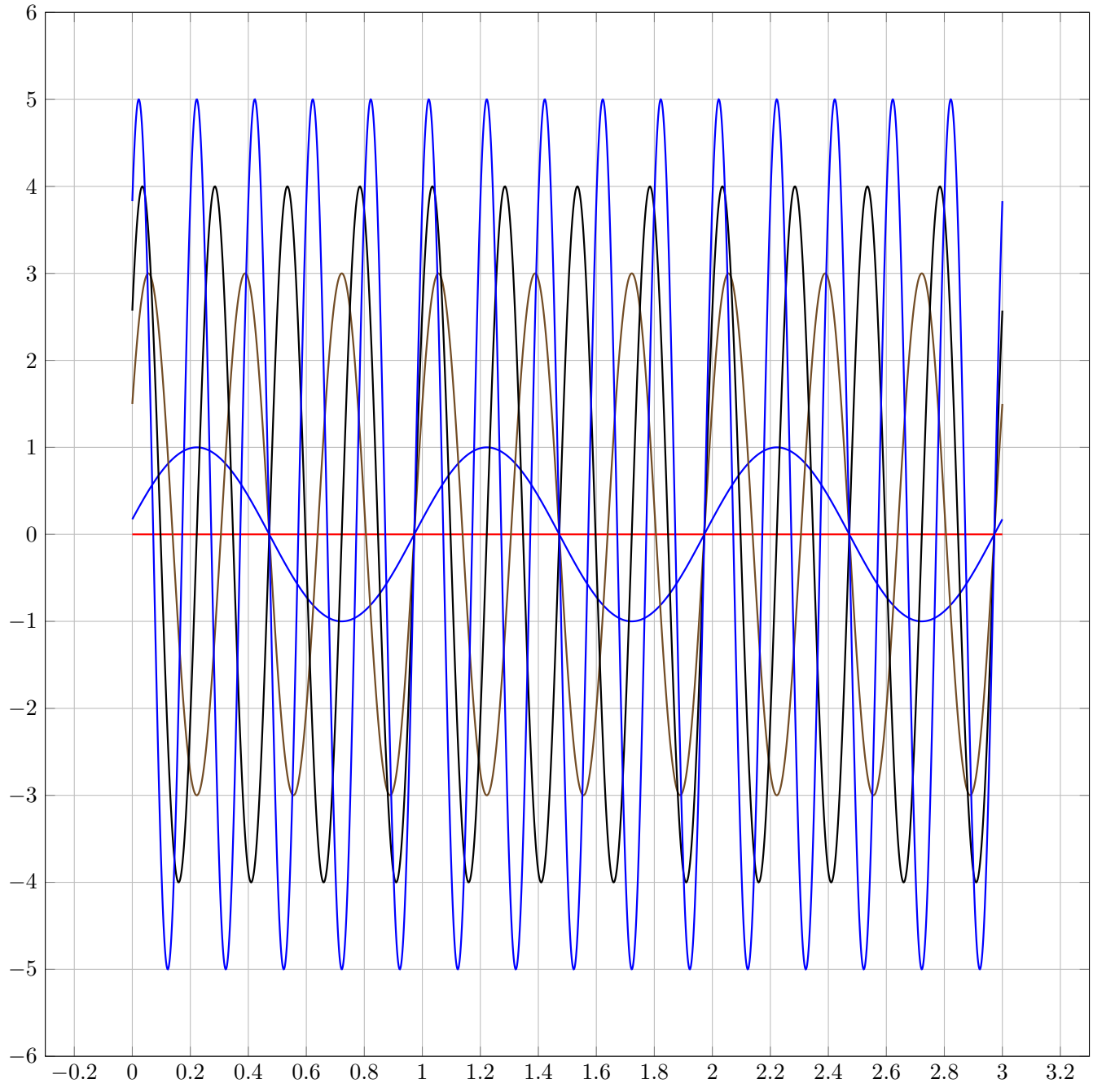
The figures below show images exported from ViVa in single strip and multiple strip modes. The ordering of waveforms in the strips can be arbitrary. The waveforms can be moved around.





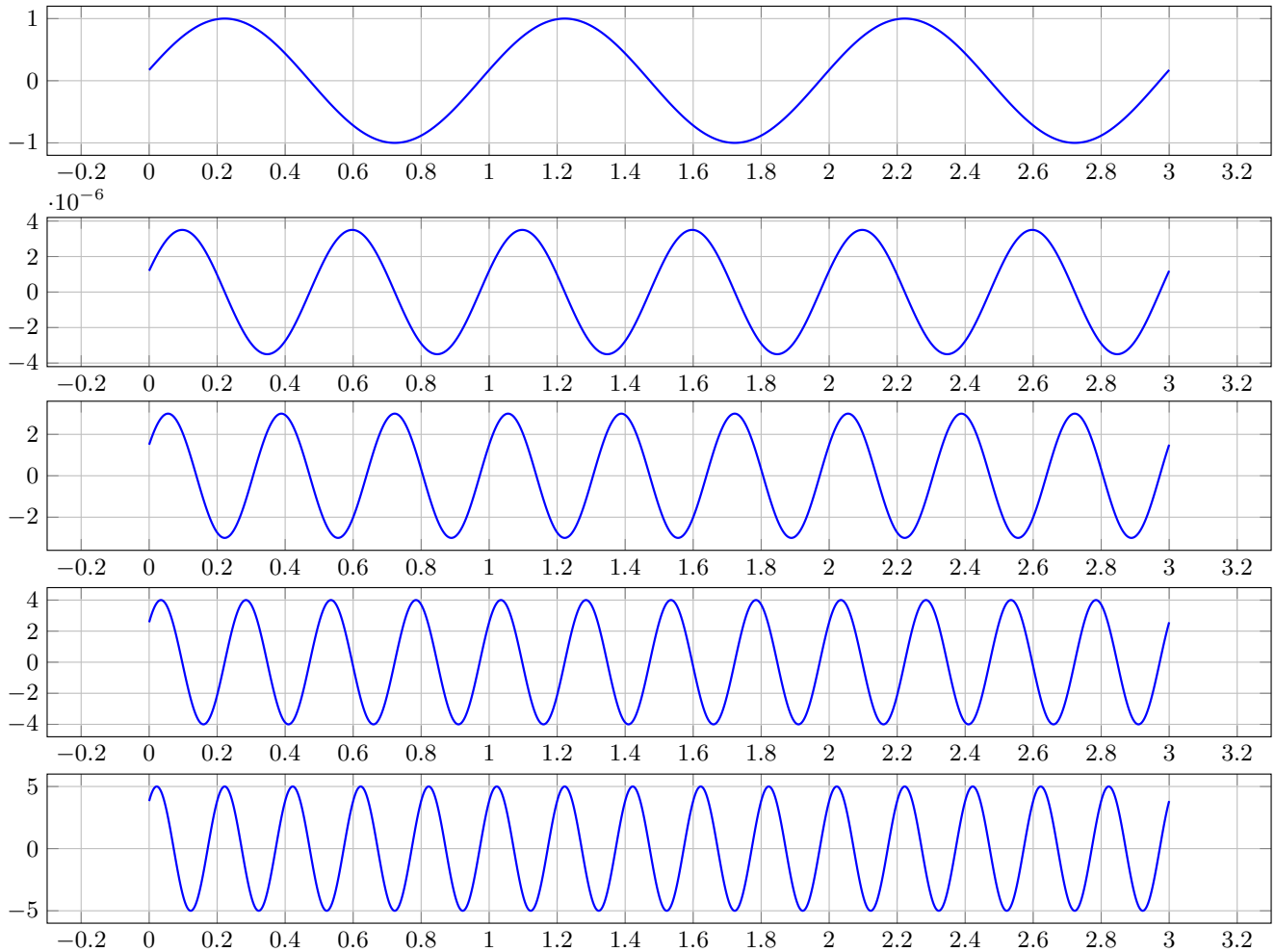
With no arguments

nkWaveToTikz()



Waveforms on different strips in a column

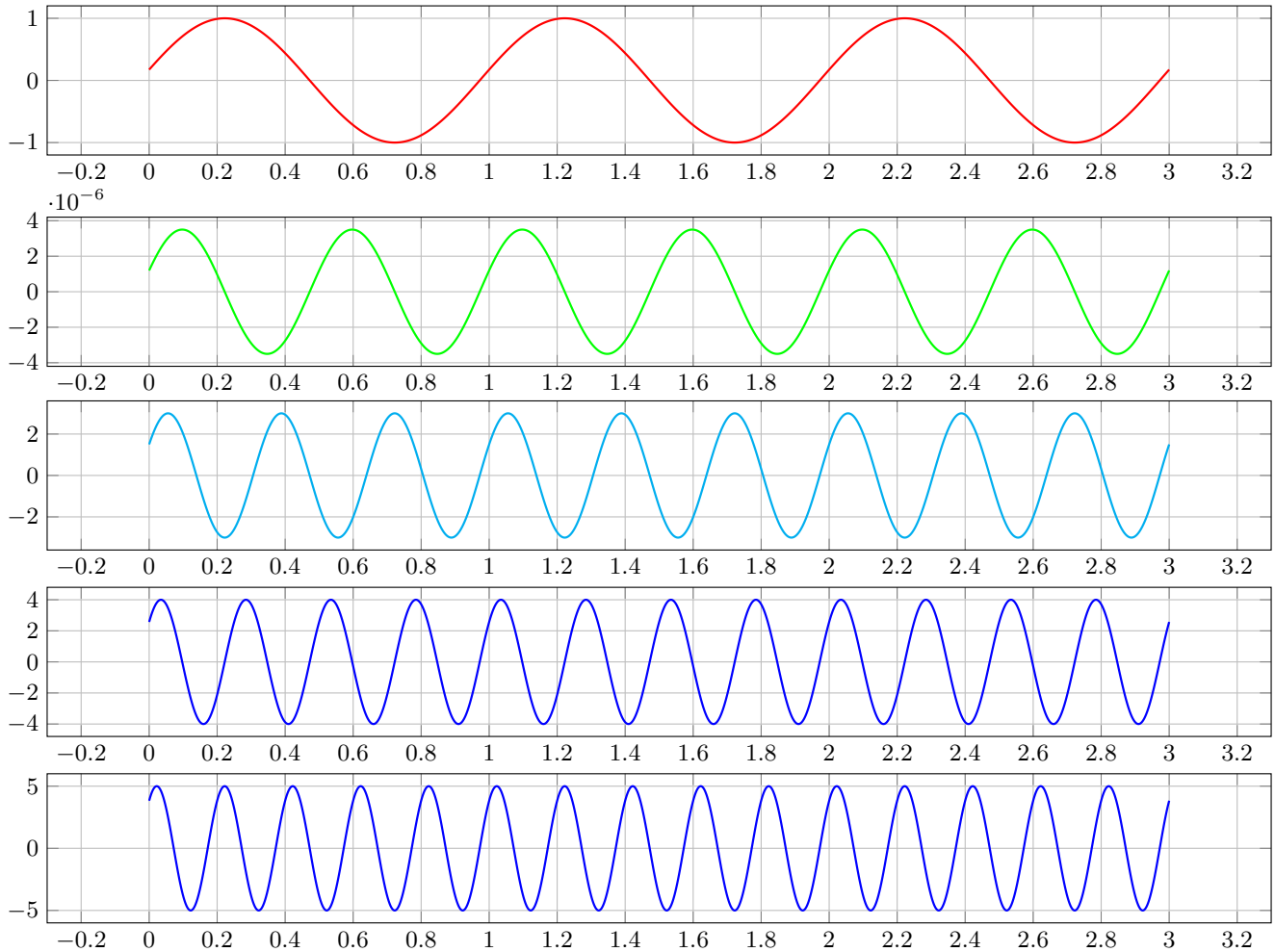
nkWaveToTikz(?plotMode "column")



Specifying waveform colors

nkWaveToTikz(?colors '(red green cyan) ?plotMode "column")

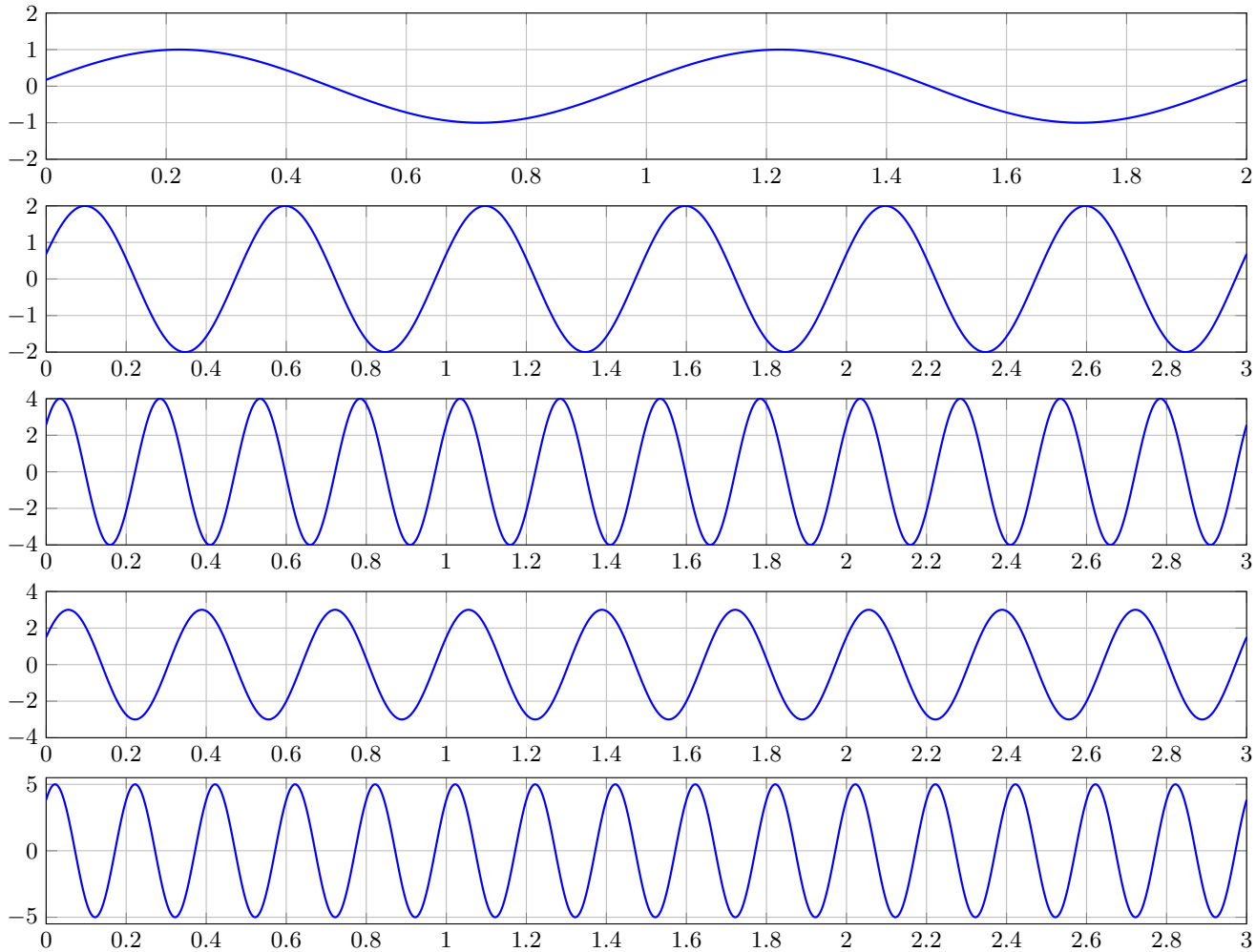
Specify waveform colors as a list. If the list has fewer items than the number of waveforms, colors are auto assigned for the rest. In the example above, the first three colors are specified and the last two are auto assigned. See this link for more information on tikz/pgf colors and the default order.



Set axis limits-can be different for each strip

nkWaveToTikz(?plotMode "column" ?xLimits '((0 2) (0 3) (0 3) (0 3) (0 3)) ?yLimits '((-2 2) (-2 2) (-4 4) (-4 4) (-5.5 5.5)))

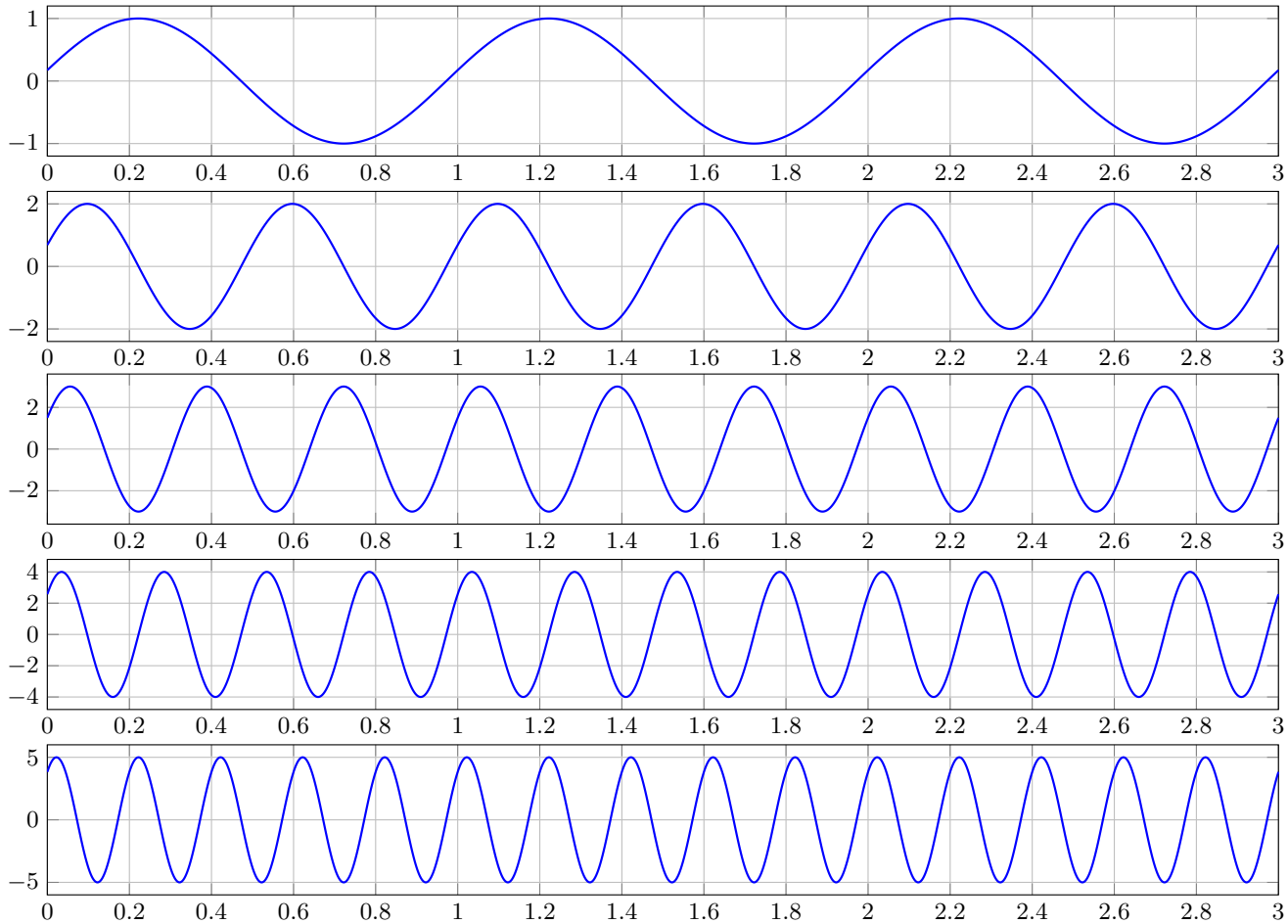
The opening parentheses of the list must be preceded by an apostrophe. The limit for each strip is consists of a list with the minimum and maximum limit.



Reorder the waveforms

For this, run the script with only `?plotMode "column"` as the argument once. In the resulting plot (e.g., previous plots), waveforms are numbered from top to bottom starting from 1. Specify the required order in `_waveAssign`.

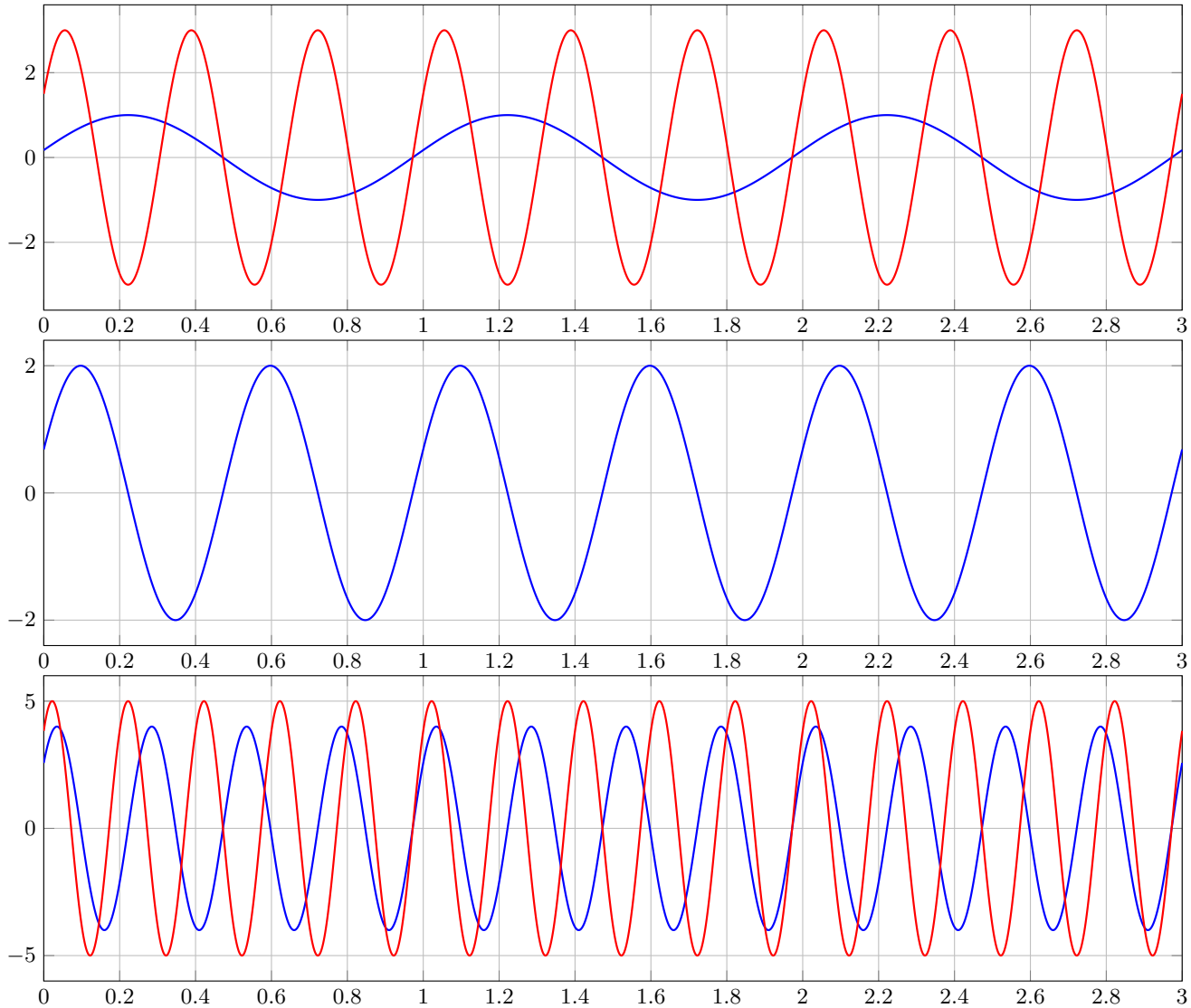
nkWaveToTikz(?plotMode ”column” ?waveAssign ’((1) (2) (4) (3) (5)) ?xLimits ’((0 3) (0 3) (0 3) (0 3) (0 3)))



Assign multiple waveforms to a strip

For this, run the script with only `?plotMode "column"` as the argument once. In the resulting plot, waveforms are numbered from top to bottom starting from 1. Specify the required order in `_waveAssign`.

nkWaveToTikz(?plotMode "column" ?waveAssign '((1 4) (2) (3 5)) ?xLimits '((0 3)))



You can also use this option to plot a single waveform in multiple strips at different zoom levels. Set plotMode to "column", have the same waveform number for all the strips in waveAssign, and axis limits as required.

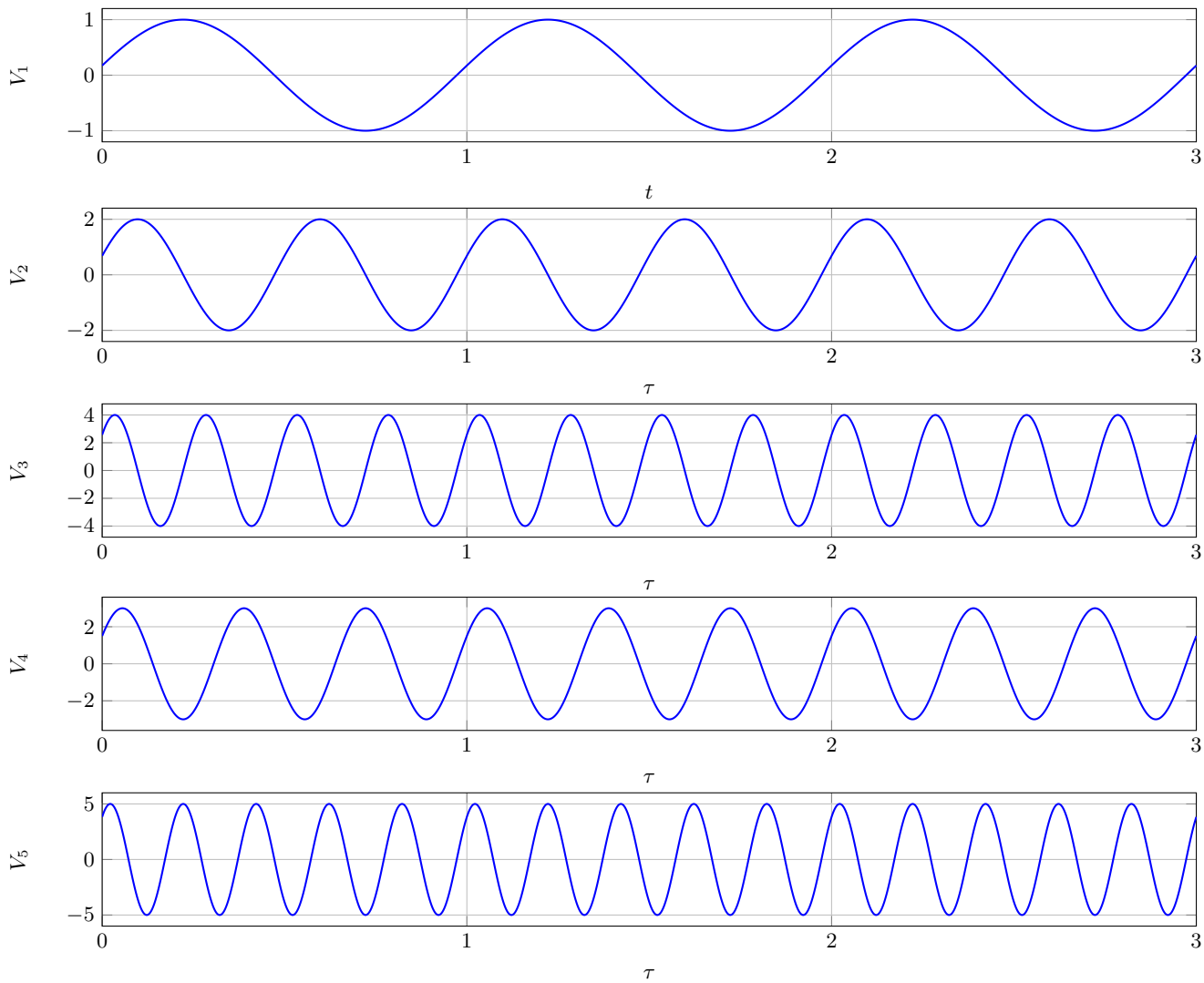
Set axis labels and tick locations

Standard LaTeX formatting works for axis labels. You have to escape some characters with an additional backslash. e.g., to get the greek letter α, use "$\\alpha$".

nkWaveToTikz(?plotMode "column" ?xLimits '((0 3) (0 3) (0 3) (0 3) (0 3)) ?xLabels '("t", "τ") ?yLabels '("V_1", "V_2", "V_3", "V_4", "V_5") ?xTicks '((0 1 2 3)))

In the above example, since only two items are in xLabel, the second one is repeated for all the later ones.

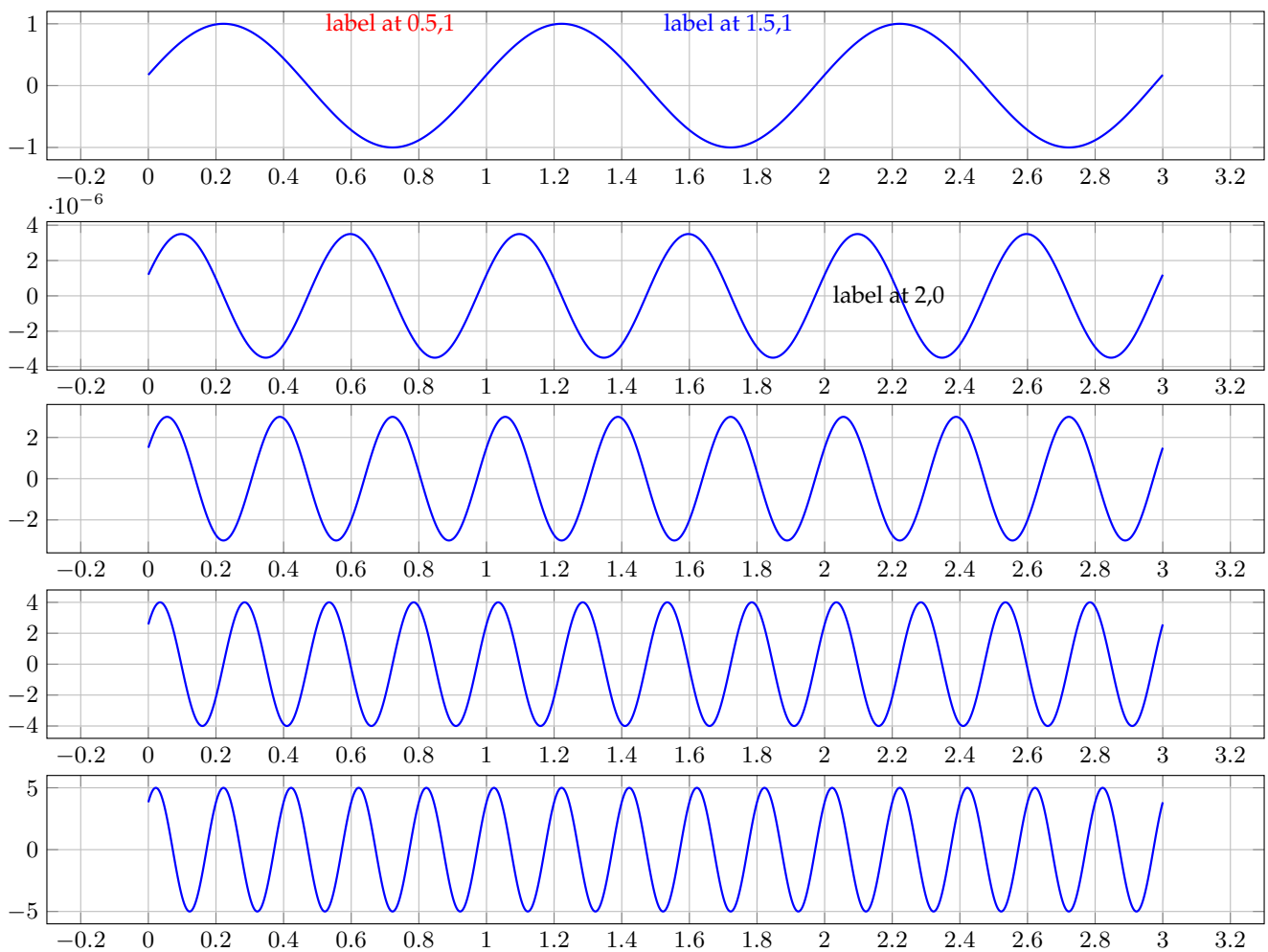
x-axis ticks and gridlines appear only at the specified points.



Add text labels on the plots

nkWaveToTikz(?fileName "withtextlabels.tikz" ?plotMode "column" ?textLabels '(((0.5 1 "label at 0.5,1" "red") (1.5 1 "label at 1.5,1" "blue")) ((2 0 "label at 2,0"))))

`_textLabels` is a list with as many items as the number of lists or fewer. Each of those lists contains a list of labels. Each list is of the form (x, y, "label text" "color"). The labels are {left, middle} aligned at the specified coordinates. "color" is optional.



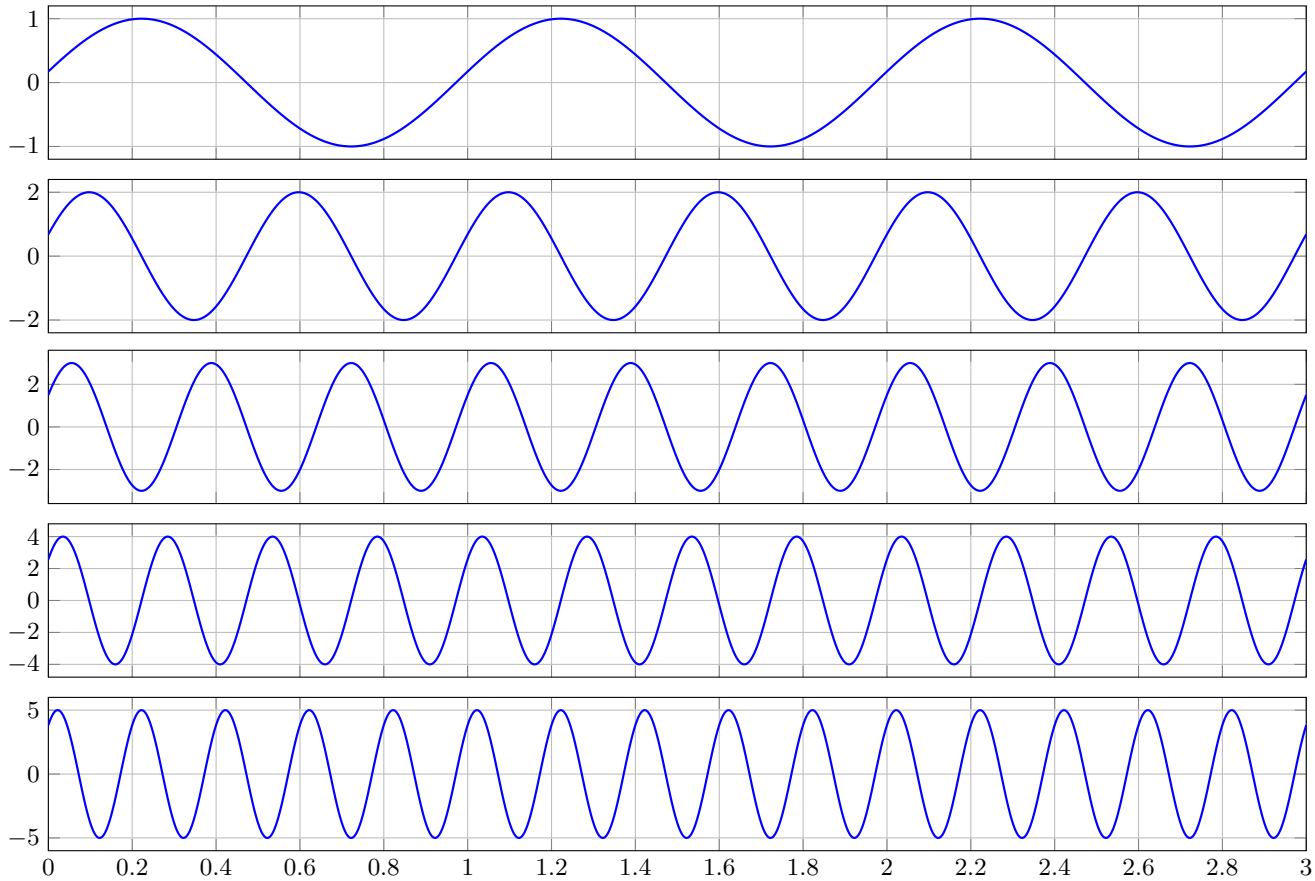
Remove numbers along the axis for compactness

When the x-axis is the same for all strips, only the bottom one needs the axis numbering. Sometimes only the shape needs to be shown with no markings on the axis.

nkWaveToTikz(?plotMode "column" ?waveAssign '((1) (2) (4) (3) (5)) ?xLimits '((0 3) (0 3) (0 3) (0 3) (0 3)) ?xTickLabels '(0 0 0 0 1))

If `_xLimits` consists of a single two-element list, it will be repeated for all the strips.

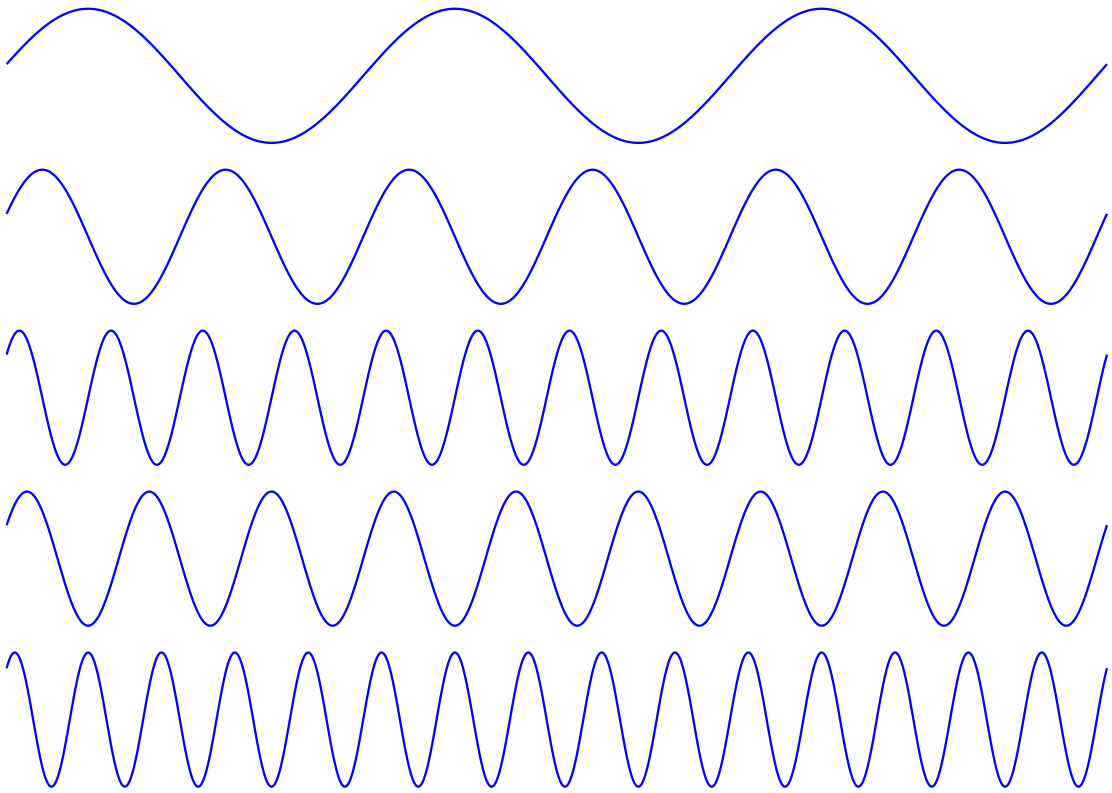
nkWaveToTikz(?plotMode "column" ?waveAssign '((1) (2) (4) (3) (5)) ?xLimits '((0 3)) ?xTickLabels '(0 0 0 0 1))



Waveforms without axes

nkWaveToTikz(?plotMode "column" ?noAxes 1)

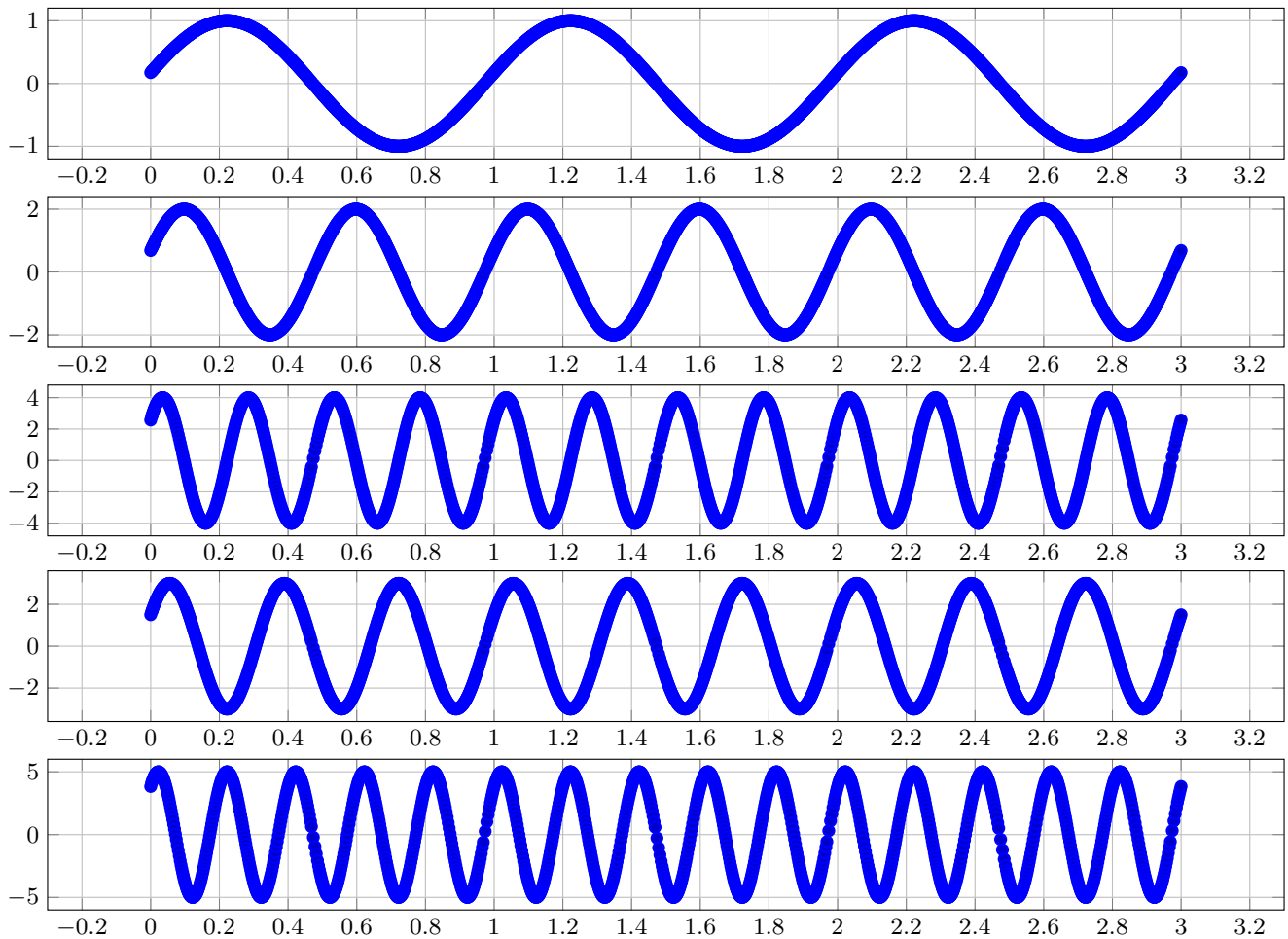
This can be useful for getting waveshapes to show on block diagrams or schematics.



Waveforms with markers

nkWaveToTikz(?plotMode "column" ?markers 1)

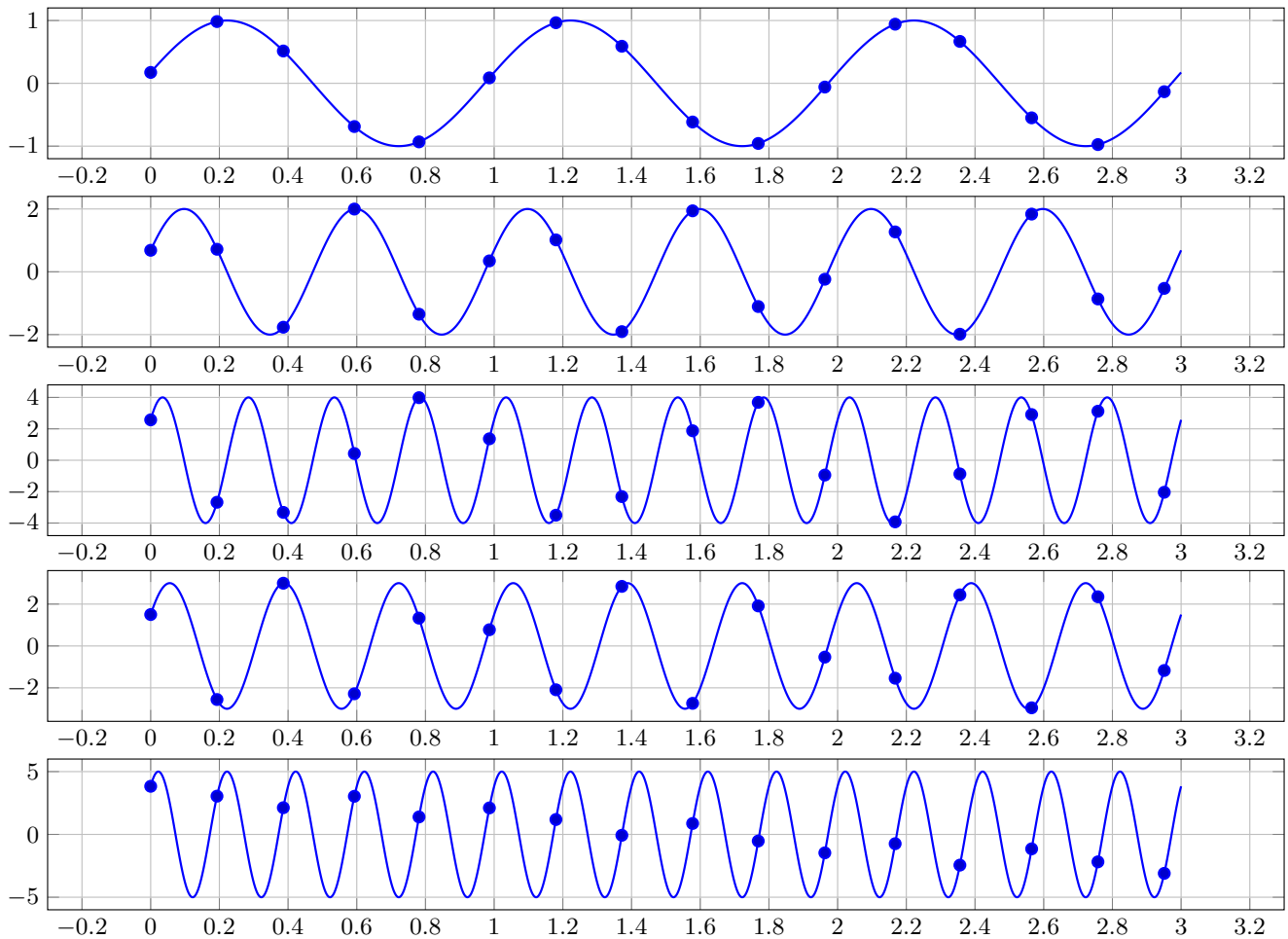
If there are a lot of points, the markers make the line very thick.



Waveforms with less frequent markers

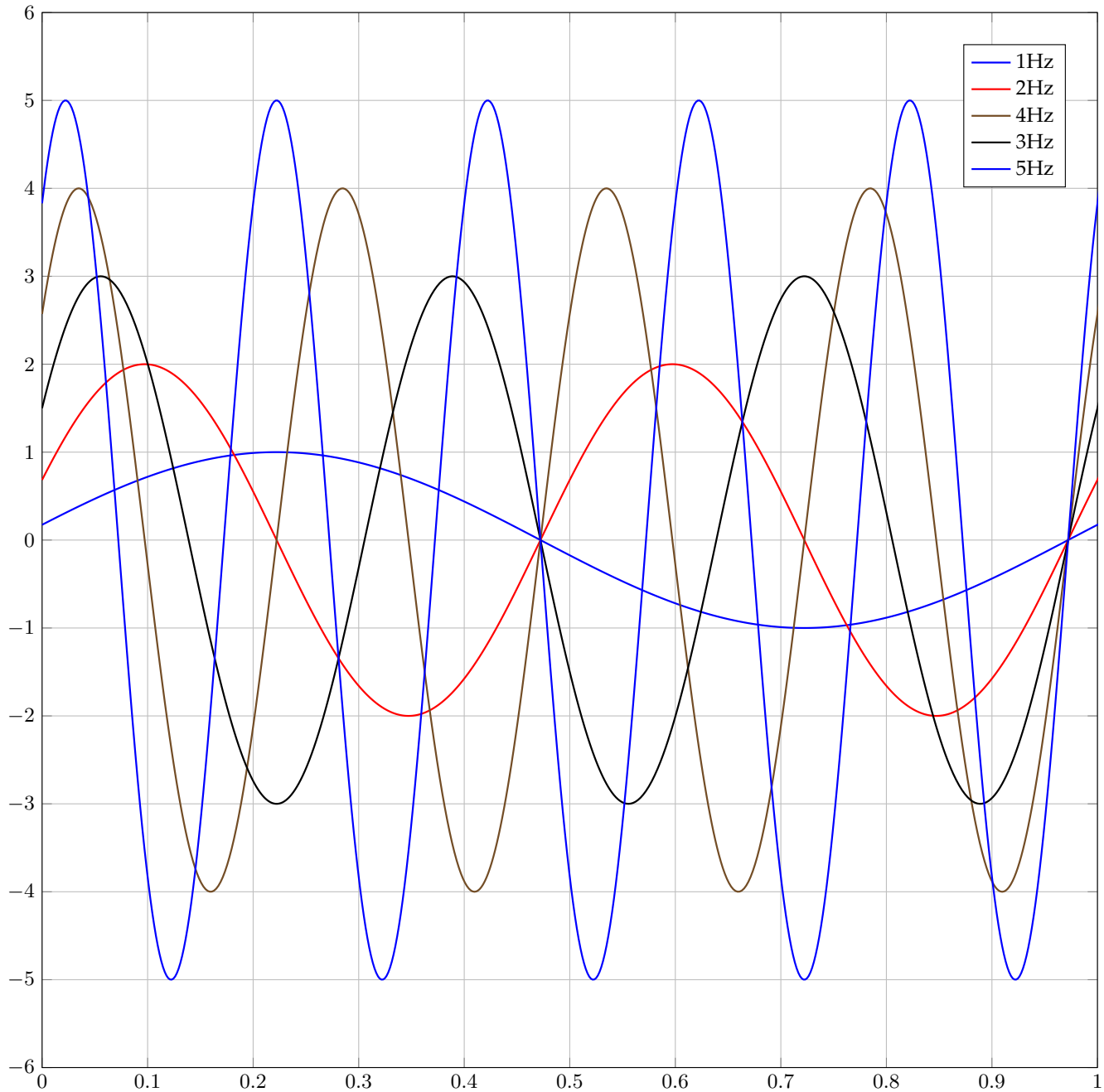
nkWaveToTikz(?plotMode "column" ?markers 100)

Markers every 100 points. They may not be uniformly spaced because the transient analysis time steps may not be uniform.



Waveforms with legends

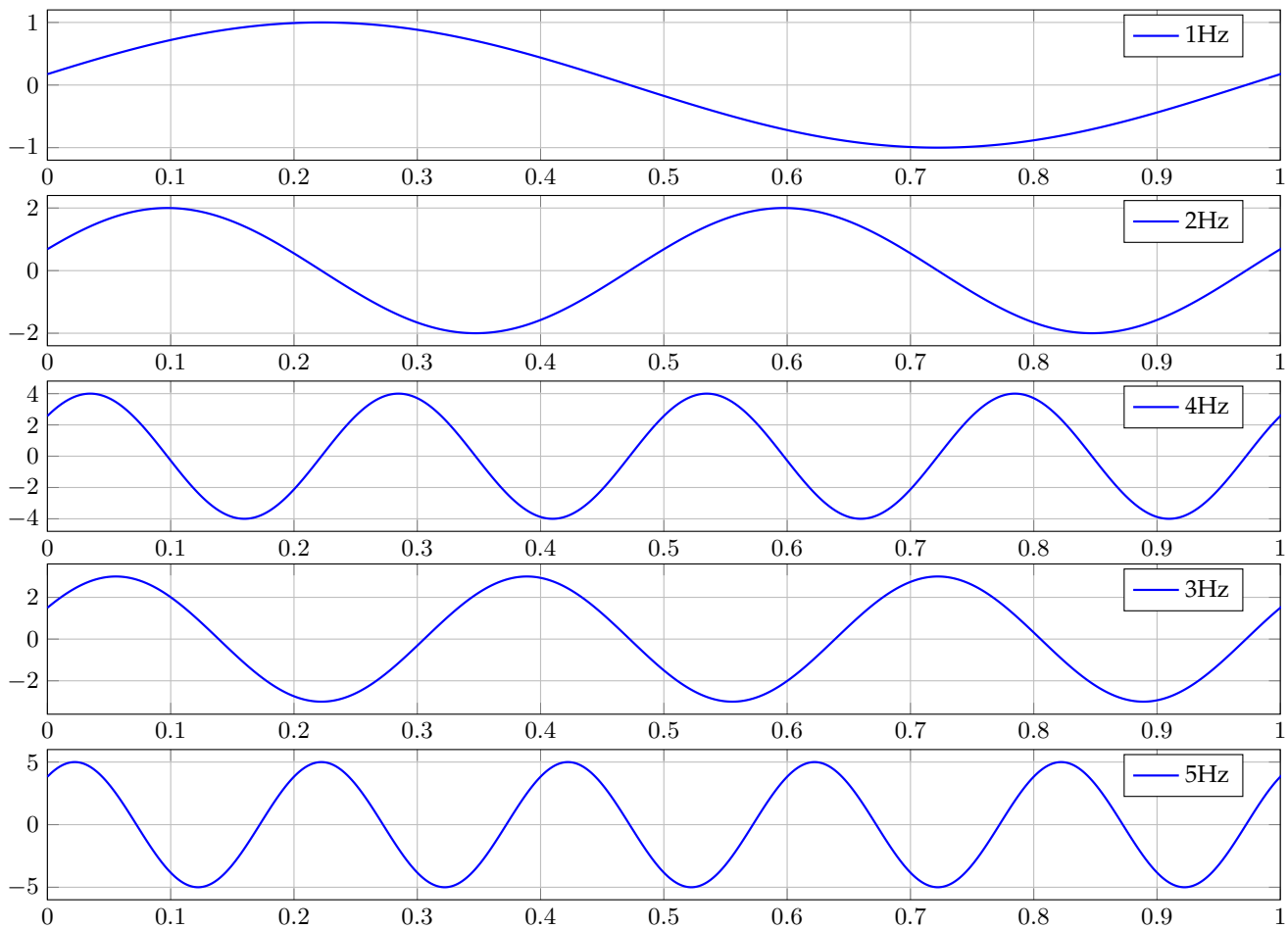
nkWaveToTikz(?xLimits '((0 1)) ?legends '("1Hz" "2Hz" "4Hz" "3Hz" "5Hz"))



Multi-strip waveforms with legends

nkWaveToTikz(?plotMode "column" ?xLimits '((0 1)) ?legends '("1Hz" "2Hz" "4Hz" "3Hz" "5Hz"))

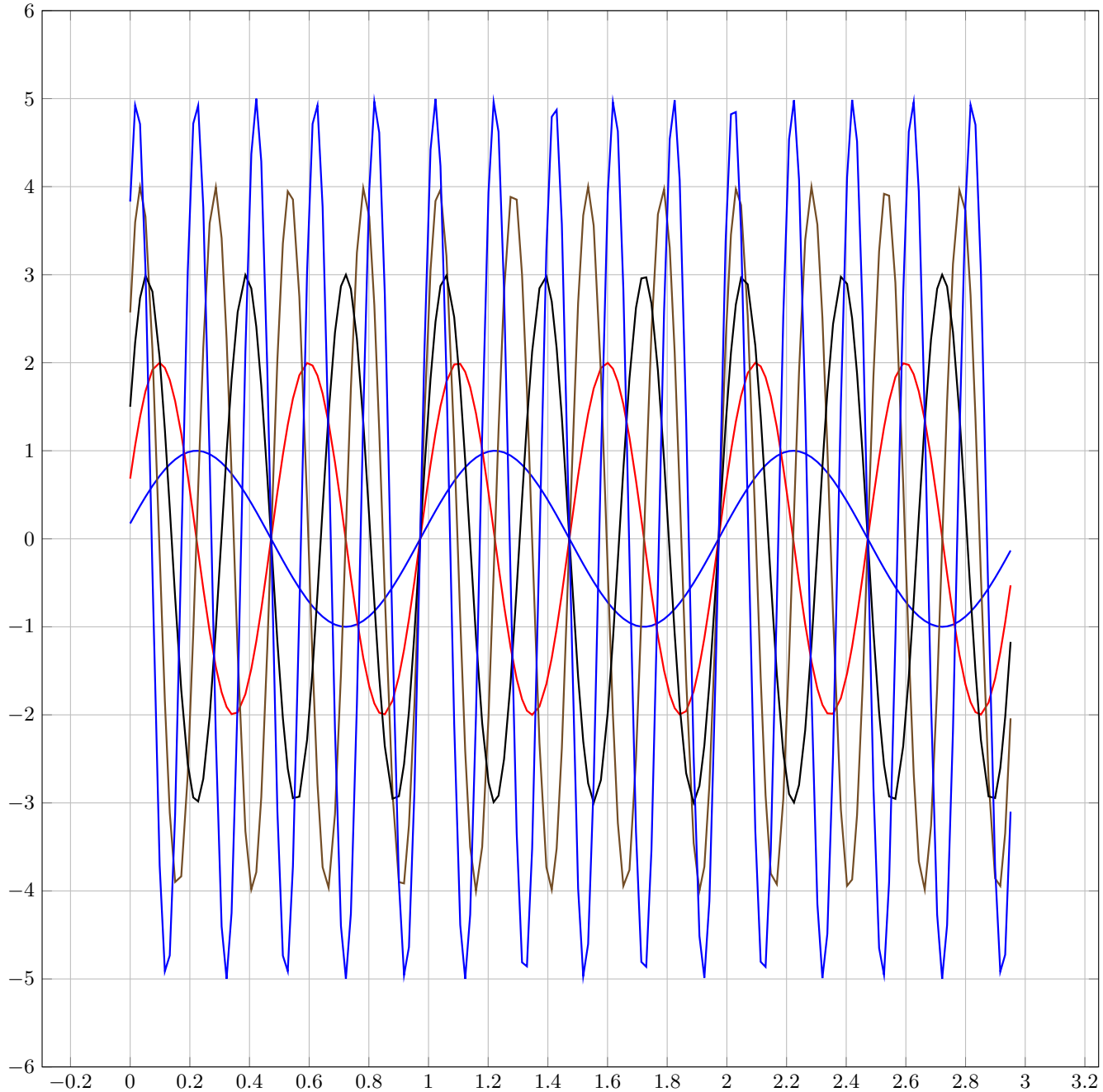
When "column" mode is selected, using legends throws an error about nested matrices, but the legends seem to come out correctly.



Undersampling the waveforms

nkWaveToTikz(?undersample 10)

The TikZ file size is nearly in inverse proportion to the undersampling factor. Use the largest undersampling factor that shows the waveform to your satisfaction. In the plot below, the lowest frequency waveform is smooth enough, but the highest frequency waveform is not.



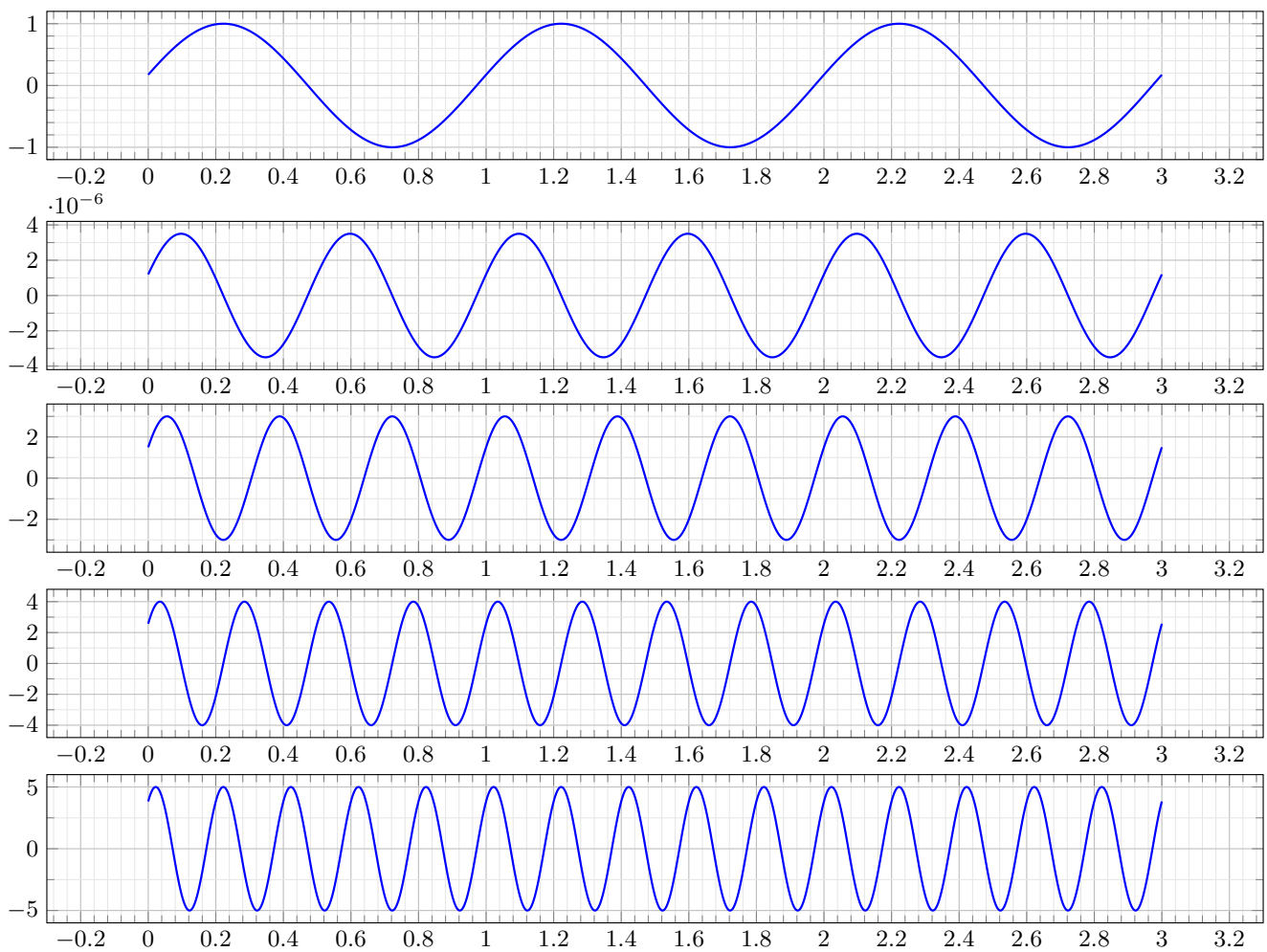
Minor grids

nkWaveToTikz(?minorGrid '((4 4) (4 1)))

List of two-element lists. For each strip, the first and second elements are the number of minor gridlines between two major gridlines on the x-axis and y-axis respectively.

In this example, the first strip has four minor grids between two major grids along both x and y axes and the other strips have four minor grids between two major grids along the x axis and one minor grid between two major grids along the y axis.

By default the minor grid lines are thinner than major grids. This can be customized specifying `additionalAxisArgs`

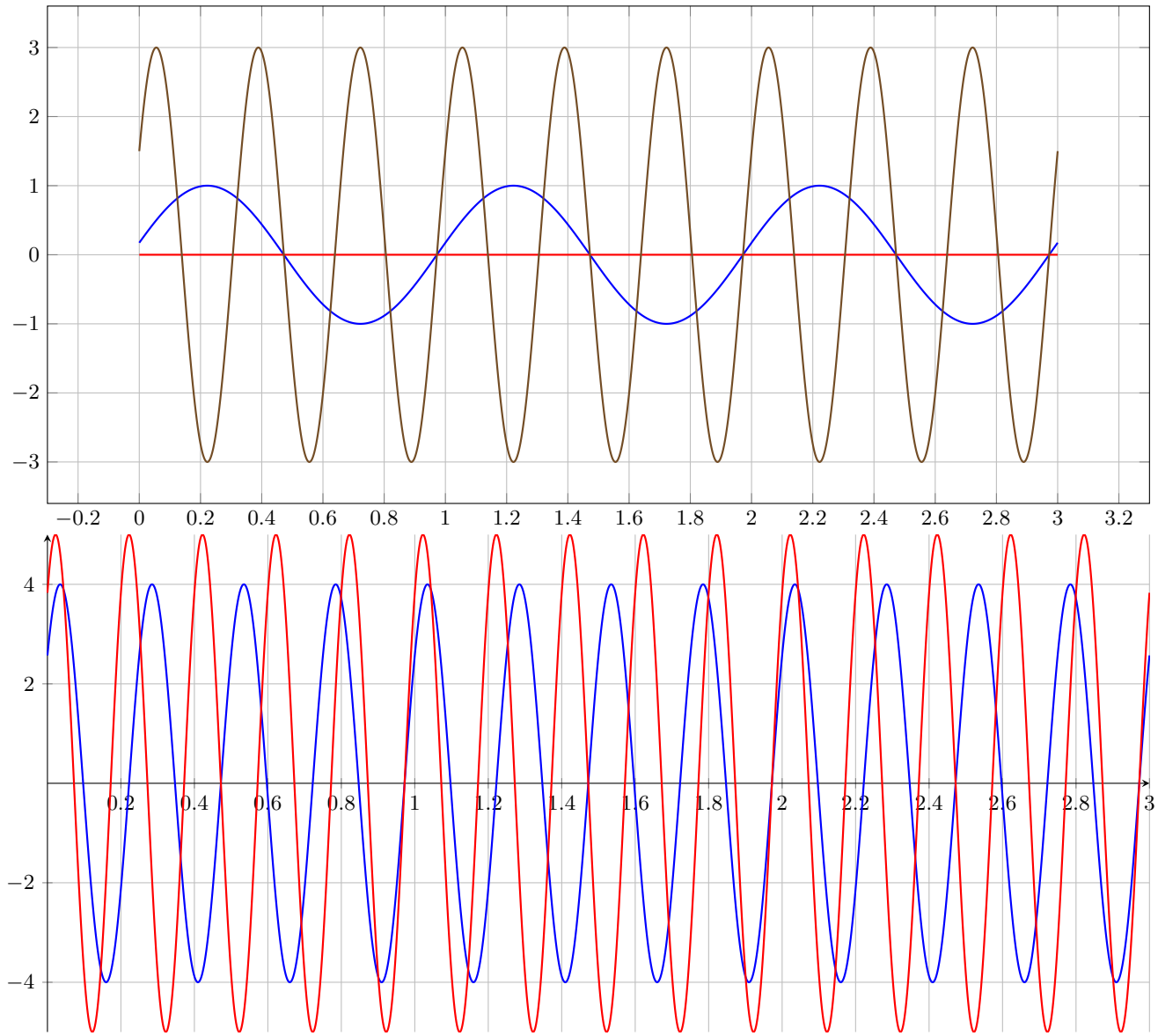


Additional axis arguments

You can specify any additional arguments (one per strip, as a single string) that will be added as the last line of the `option`for \begin{axis}[`options`]. The string can have multiple comma-separated options. If the option has been specified earlier, it will be superseded.

nkWaveToTikz(?additionalAxisArgs '(nil "axis lines=center") ?waveAssign '((1 2 3) (4 5)) ?plotMode "column")

In this example, the waveforms are assigned to two strips. The first strip has the default axis lines (left) and the second uses "axis lines=center".



Generating a pdf of the figure

nkWaveToTikz(?genPdf 1)

Using the option above, a `pdf` cropped to the dimensions of the plot can be generated which can then be included using `includegraphics`. For this to work, the file `headerFile.tex` has to be in the folder specified by `headerFilePath`. If `headerFilePath` is not specified `headerFile.tex` has to be in the virtuoso startup folder. The header file can be edited to modify the appearance. The script uses `pdflatex`.

There are two files `headerFileCropped.tex` and `headerFileFullpage.tex`. These generate cropped pdfs and full page pdfs, respectively. The file headerFile.text is a copy of the former. Copy the latter if full page pdfs are required. The former uses `minimal` environment with `preview` package. The latter uses `extarticle` environment. These files can be edited as required.

The script runs `pdflatex` with the option `-interaction=nonstopmode` so that it completes without waiting for user input. All the intermediate files (`tex`, `aux`, `log`) are deleted. Errors in the LaTeX file can result in no `pdf` being generated. For debugging such cases, set `retainTexFile` to 1. This leaves the tex file which you can run locally. `headerFile.tex` can be modified suitably after debugging.

nkWaveToTikz(?genPdf 1 ?retainTexfile 1)

Pros and cons of using `tikz` and `pdf` are listed below.

Using the `tikz` file:

+ The TikZ file will be compiled along with the LaTeX document and will have the same fonts and other settings.
+ Does not need a local LaTeX installation.
− The figure will be processed with LaTeX. This can get very slow with a large number of figures.
− Can be used only with LaTeX documents.

Using the `pdf` file:

+ Compilation is faster than with `TikZ` figures.
+ Can be used outside LaTeX documents. `pdf` can be converted to, e.g., `png` using ghostscript.
− Differences in LaTeX settings between those used for pdf generation and the final LaTeX document can cause mismatch in style between the document and the figure.
− Needs a local LaTeX installation with all the desired packages.

Using the TikZ plots in your LaTeX document

You need the following lines in your document header. See the file testTikz.tex in the same folder for an example.

\usepackage{pgf,pgfarrows,pgfnodes,pgfautomata,pgfheaps,pgfshade}

\usepackage{tikz}

You can include the the plots using "\input". Example below. "\resizebox" can be used to scale the picture as required.

\resizebox{0.5\textwidth}{!}{\input{tikZFileName.tikz}}

The TikZ file can be edited as required to customize the picture (color, marker, legend, any other formatting).

The TikZ file can be included in a LaTeX file and the compiled pdf cropped and exported (using gs) and used with \includegraphics.

If you want to change all font sizes in the plot, enclose the \input line within \begin{large} . . . \end{large} or any other size that you prefer. e.g.,

\resizebox{0.5\textwidth}{!}{\begin{large} \input{tikZFileName.tikz} \end{large} }